

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Eksamen

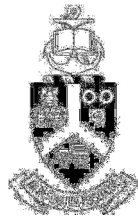
Kopiereg voorbehou

Examination

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Module EBN111
Elektrisiteit en Elektronika
15 Junie 2015

Module EBN111
Electricity and Electronics
15 June 2015



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting: Exam information:

Maksimum punte: Maximum marks:	90	Volpunte: Full marks:	90
Duur van vraestel: Duration of paper:	180 minute 180 minutes	Oopboek / toeboek: Open / closed book:	Toe Closed
Totale aantal bladsye (hierdie blad ingesluit): Total number of pages (including this page):		29	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Polêre vorm: $A \angle \theta$; Reghoekige vorm: $a + jb$
Polar form: $A \angle \theta$; Rectangular form: $a + jb$
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): Lecturer(s):	J. Joubert
	A. Chatterjee
Eksterne eksaminator: External examiner:	J.W. Odendaal

VRAAG/QUESTION 1 [45]

Vraag 1.1 [1]

Bereken die lading verteenwoordig deur 3 miljoen protone. Gee jou antwoord in pico-Coulomb (pC).

Question 1.1 [1]

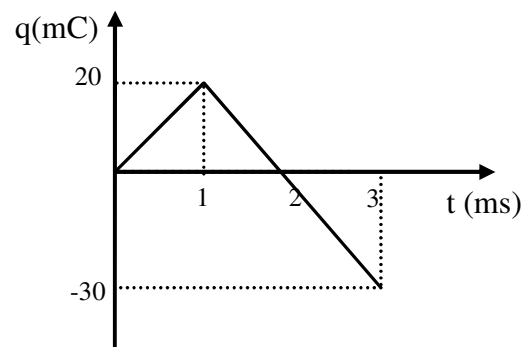
Calculate the charge represented by 3 million protons. Give your answer in pico-Coulomb (pC).

Vraag 1.2 [2]

Bepaal die stroom by 2ms.

Question 1.2 [2]

Find the current at 2ms.cvv



Vraag 1.3 [2]

Bereken die energie (in Joule) wat nodig is om 2C lading in 3s deur 'n 5Ω weerstand te beweeg.

Question 1.3 [2]

Find the energy (in Joule) required to move 2C of charge through a 5Ω resistor in 3s.

Vraag 1.4 [2]

'n Elektriese motor kan 50kJ lewer in 2 minute. Bereken die jaarlikse koste (in Rand) as die motor vir 5 ure 'n dag, 300 dae 'n jaar, gebruik word. Die koste van elektrisiteit is R1/kWh.

Question 1.4 [2]

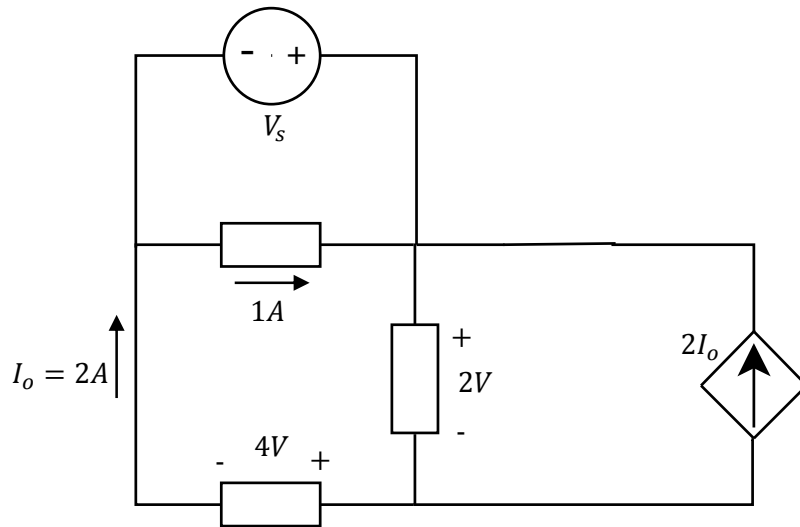
An electric motor can deliver 50kJ in 2 minutes. Find the annual cost (in Rand) of running the motor if it is used for 5 hours a day, 300 days a year. The cost of electricity is R1/kWh.

Vraag 1.5 [4]

- a) Bereken die drywing ge-assosieer met die afhanklike stroombron.
- b) Bereken die drywing ge-assosieer met die onafhanklike stroombron.

Question 1.5 [4]

- a) Find the power associated with the dependent current source.
- b) Find the power associated with the independent voltage source.



Vraag 1.6 [2]

'n 5cm lengte silicondraad het 'n geleidingsvermoë van 1×10^{-4} S. Bereken die deursnitoppervlak (in m^2) van die draad, as die resistiwiteit (ρ) van silicon $6.4 \times 10^2 \Omega\text{m}$ is.

Question 1.6 [2]

A 5cm long piece of silicon wire has a conductance of 1×10^{-4} S. Find the cross-sectional area (in m^2) of the wire, if the resistivity (ρ) of silicon is $6.4 \times 10^2 \Omega\text{m}$.

Vraag 1.7 [2]

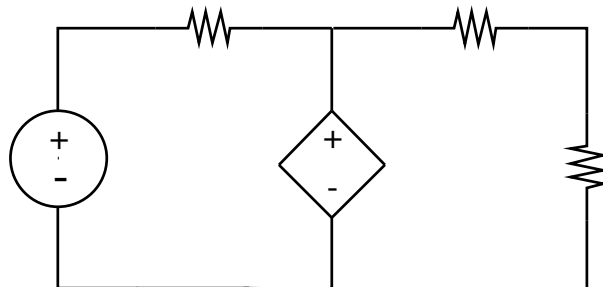
Bepaal:

- a) Die aantal nodes.
- b) Die aantal takke.

Question 1.7 [2]

Find:

- a) The number of nodes.
- b) The number of branches.



Vraag 1.8 [4]

Bepaal die ekwivalente weerstand soos gesien deur terminale a-b as:

a) $R = 60\Omega$.

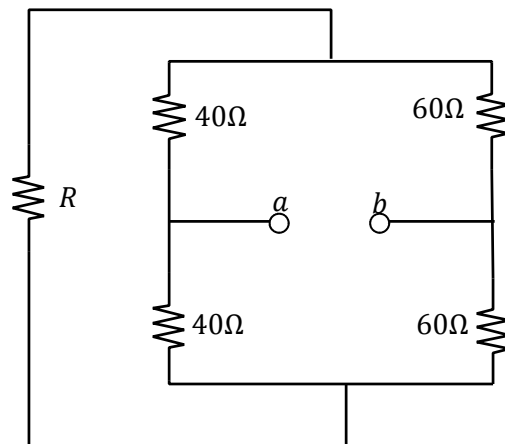
b) $R = \infty\Omega$.

Question 1.8 [4]

Find the equivalent resistance as seen by terminals a-b if:

a) $R = 60\Omega$.

b) $R = \infty\Omega$.



Vraag 1.9 [3]

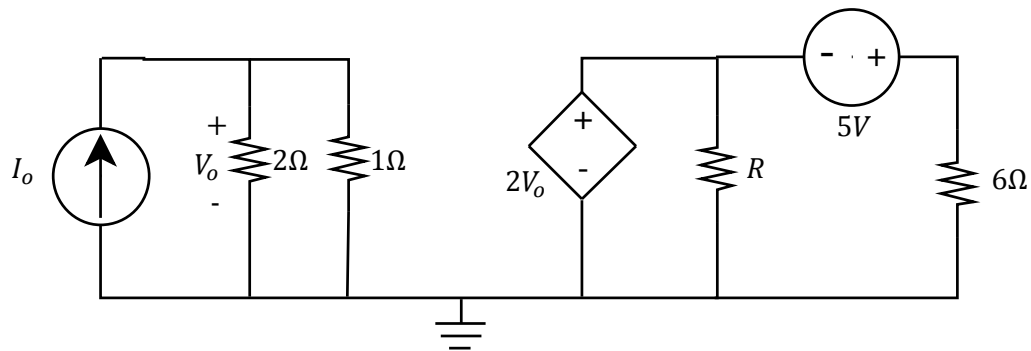
Bepaal:

- a) V_o as $I_o = 3A$
b) Die drywing ge-assosieer met die 6Ω weerstand as $V_o = 0.5V$.

Question 1.9 [3]

Find:

- a) V_o if $I_o = 3A$
b) The power associated with the 6Ω resistor if $V_o = 0.5V$.



Vraag 1.10 [4]

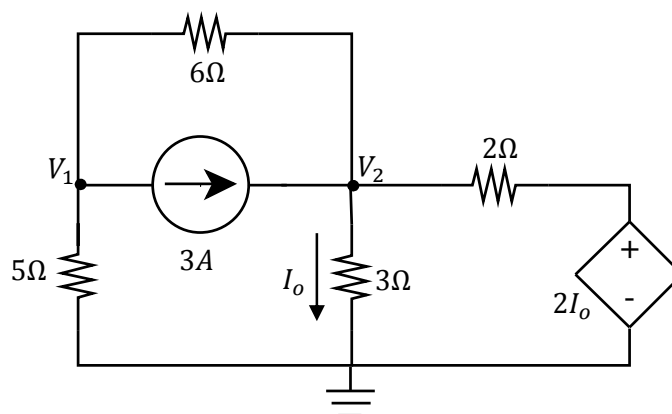
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} aV_1 - 5V_2 &= b \\ cV_1 + dV_2 &= 18 \end{aligned}$$

Question 1.10 [4]

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned} aV_1 - 5V_2 &= b \\ cV_1 + dV_2 &= 18 \end{aligned}$$



Vraag 1.11 [4]

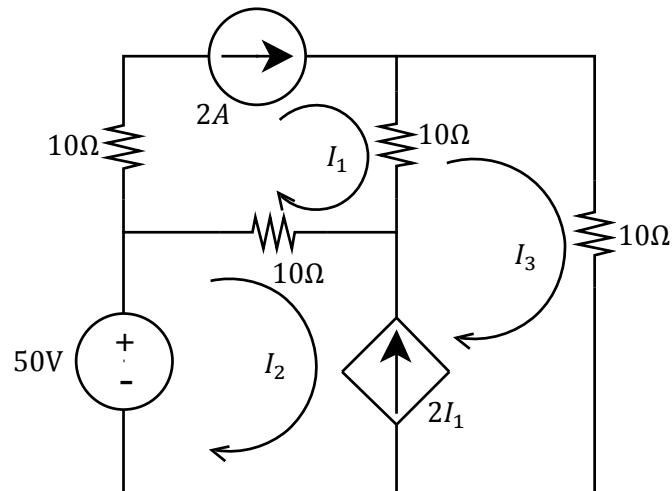
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$\begin{aligned} eI_2 + fI_3 &= 90 \\ -I_2 + gI_3 &= h \end{aligned}$$

Question 1.11 [4]

Use mesh analysis to set up two equations of the following form:

$$\begin{aligned} eI_2 + fI_3 &= 90 \\ -I_2 + gI_3 &= h \end{aligned}$$



Vraag 1.12 [4]

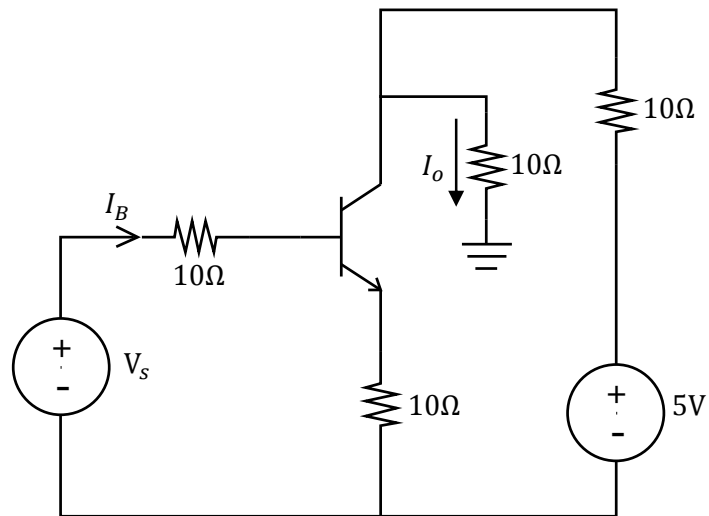
Aanvaar $V_{BE} = 0.7\text{V}$, en $\beta = 100$.

- a) Bepaal die kollektorstroom I_C (in mA) as $V_s = 2\text{V}$.
b) Aanvaar $I_B = 1.5\text{ mA}$. Bepaal die stroom I_o (in mA).

Question 1.12 [4]

Assume $V_{BE} = 0.7\text{V}$, and $\beta = 100$.

- a) Find the collector current I_C (in mA) if $V_s = 2\text{V}$.
b) Assume $I_B = 1.5\text{ mA}$. Find the current I_o (in mA).

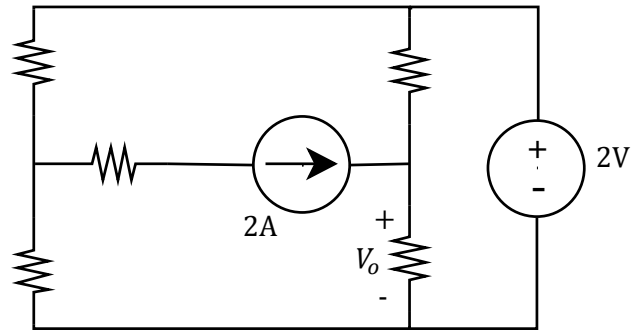


Vraag Question 1.13 [2]

Bepaal V_o slegs agv die stroombron. Alle weerstande is 10Ω .

Question 1.13 [2]

Find V_o due to the current source only. All resistor values are 10Ω .

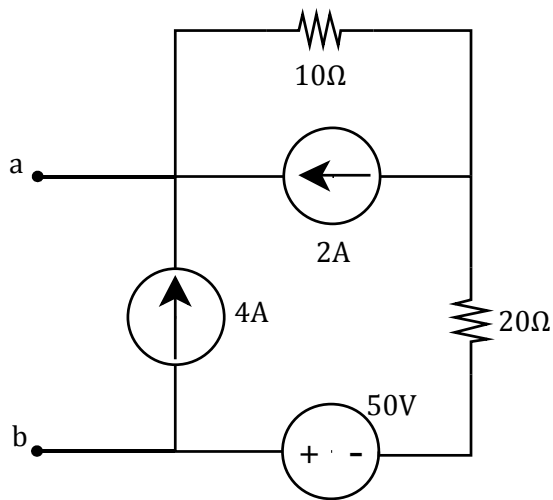
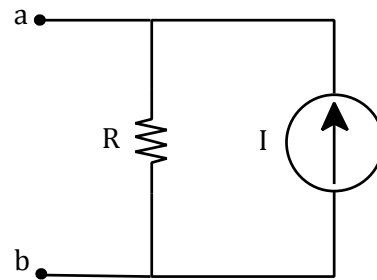


Vraag 1.14 [3]

Gebruik brontransformatie om die stroombaan van Fig. (a) te transformer na Fig. (b). Bepaal die waardes van R en I.

Question 1.14 [3]

Use source transformation to transform the circuit in Fig. (a) into that in Fig. (b). Find the values of R and I.

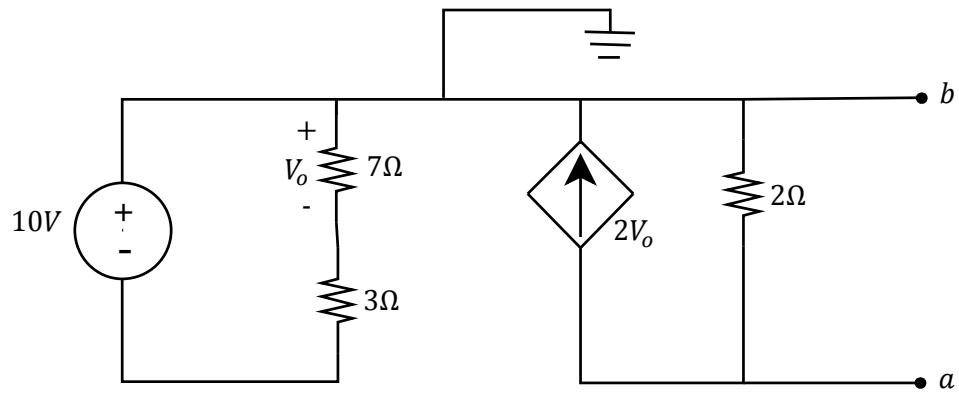
**Fig. (a)****Fig. (b)**

Vraag 1.15 [2]

Bepaal die Norton ekwivalente stroom tov terminale a-b.

Question 1.15 [2]

Find the Norton equivalent current wrt terminals a-b.

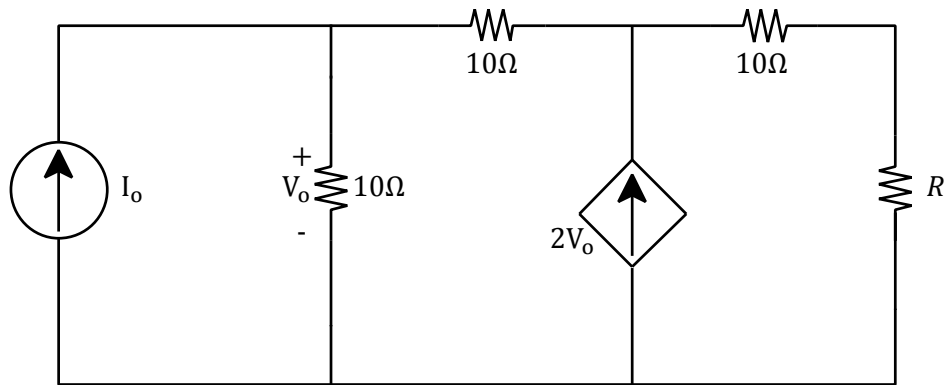


Vraag 1.16 [2]

Bepaal die waarde van R , sodat R maksimum drywing sal dissipeer.

Question 1.16 [2]

Find the value of R , such that R will dissipate maximum power.



Vraag 1.17 [2]

'n Lasweerstand in 'n stroombaan is gekies om maksimum drywing te gedissipeer. As die weerstand 'n waarde van 50Ω het en $50W$ in die weerstand gedissipeer word; bereken die Thevenin ekwivalente spanning soos gesien deur die weerstand.

Question 1.17 [2]

A load resistor in a circuit is chosen for maximum power dissipation. If the resistor has a value of 50Ω and $50W$ is dissipated by the resistor; calculate the Thevenin equivalent voltage as seen by this resistor.

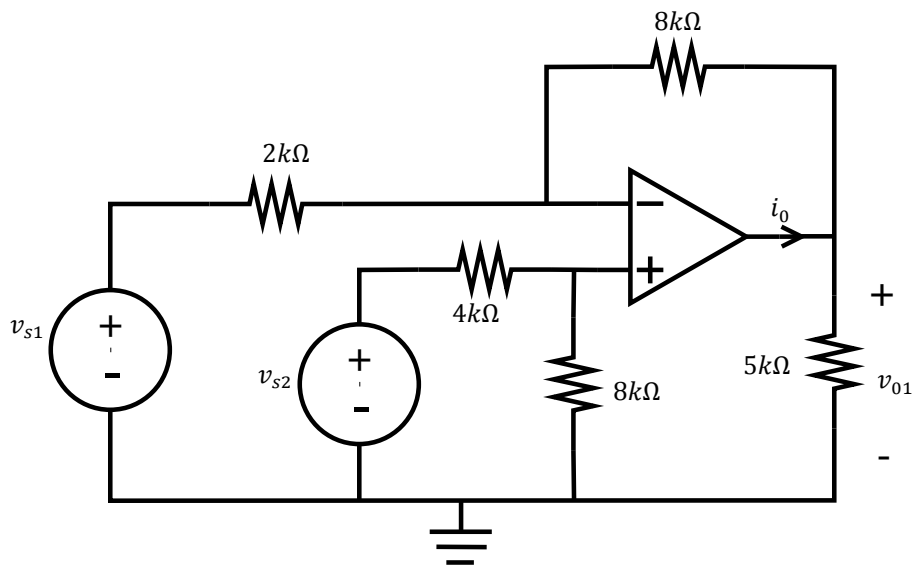
VRAAG/QUESTION 2 [23]

Vraag 2.1 [4]

- a) Bepaal v_{01} indien $v_{s1} = 3V$ en $v_{s2} = 3V$.
b) Bepaal i_0 indien $v_{s1} = 0V$ en $v_{s2} = 1.5V$.

Question 2.1 [4]

- a) Determine v_{01} if $v_{s1} = 3V$ and $v_{s2} = 3V$.
b) Determine i_0 if $v_{s1} = 0V$ and $v_{s2} = 1.5V$.

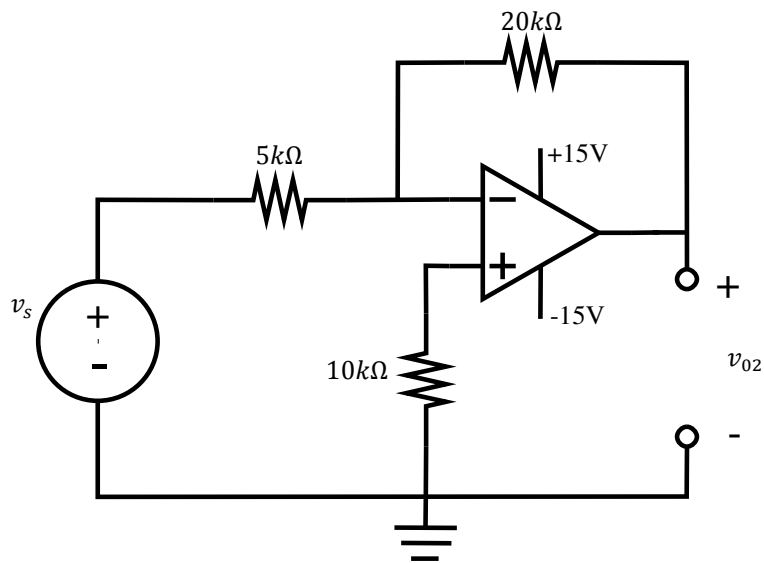


Vraag 2.2 [3]

- a) Bepaal v_{02} as $v_s = -1V$.
b) Wat is die minimum waarde van v_s voordat die OV in versadiging sal ingaan.

Question 2.2 [3]

- a) Determine v_{02} if $v_s = -1V$.
b) What is the minimum value v_s can be before the op-amp goes into saturation.

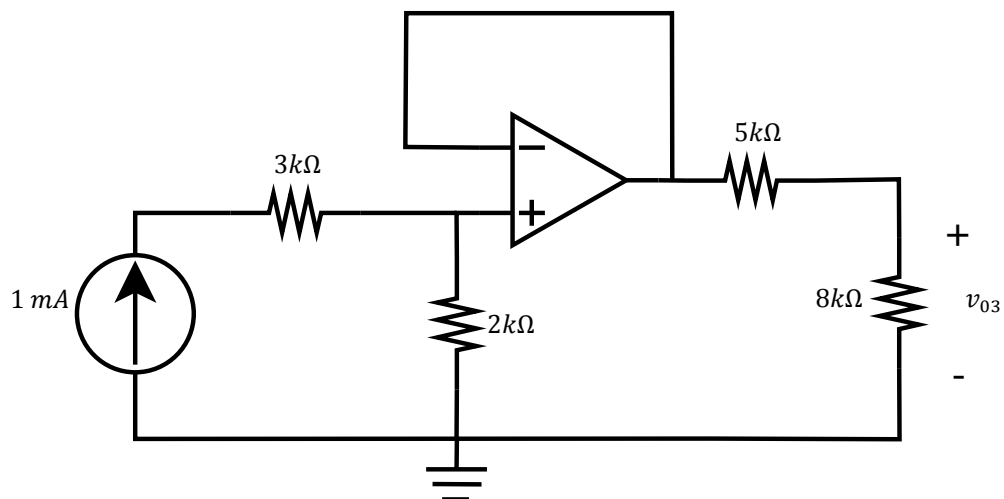


Vraag 2.3 [2]

Bepaal v_{03} .

Question 2.3 [2]

Determine v_{03} .

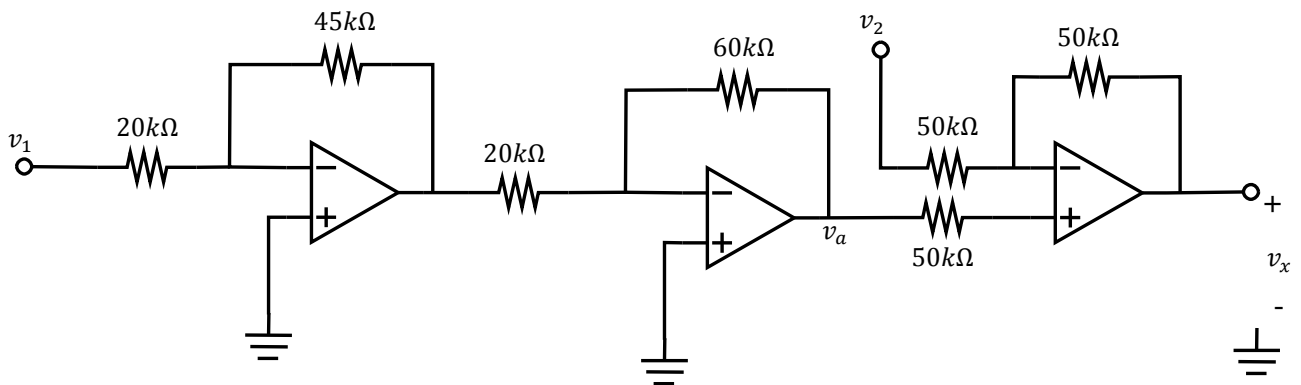


Vraag 2.4 [4]

- a) Aanvaar $v_2 = 0$, en bereken die versterkingsfaktor v_x/v_1 .
- b) Aanvaar $v_a = 3V$ en $v_2 = 2V$, en bepaal v_x .

Question 2.4 [4]

- a) Assume $v_2 = 0$, and calculate the amplification factor v_x/v_1 .
- b) Assume $v_a = 3V$ and $v_2 = 2V$, and calculate v_x .

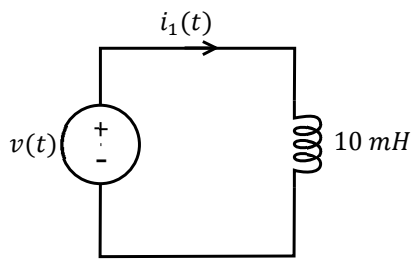


Vraag 2.5 [4]

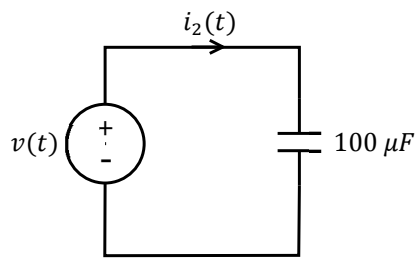
- a) Die spanning $v(t)$ word aangelê oor 'n 10 mH induktor (sien Figuur (a)). Bepaal i_1 by $t = 2\text{ s}$. (aanvaar $i_1(0) = 0$).
- b) Dieselfde spanning $v(t)$ word aangelê oor 'n $100\text{ }\mu\text{F}$ kapasitor (sien Figuur (b)). Bepaal i_2 by $t = 4\text{ s}$.

Question 2.5 [4]

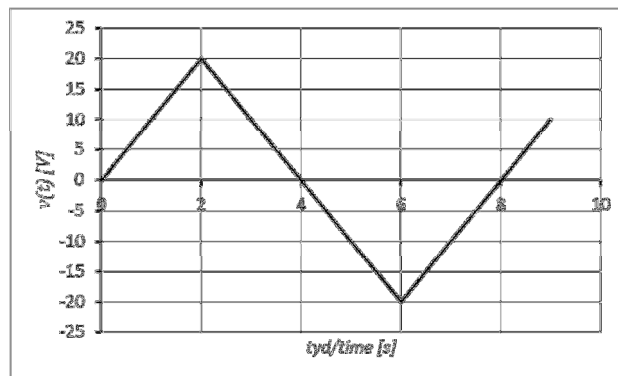
- a) The voltage $v(t)$ is applied across a 10 mH inductor (see Figure (a)). Determine i_1 at $t = 2\text{ s}$. (assume $i_1(0) = 0$).
- b) The same voltage $v(t)$ is applied across a $100\text{ }\mu\text{F}$ capacitor (see Figure (b)). Determine i_2 at $t = 4\text{ s}$.



(a)



(b)

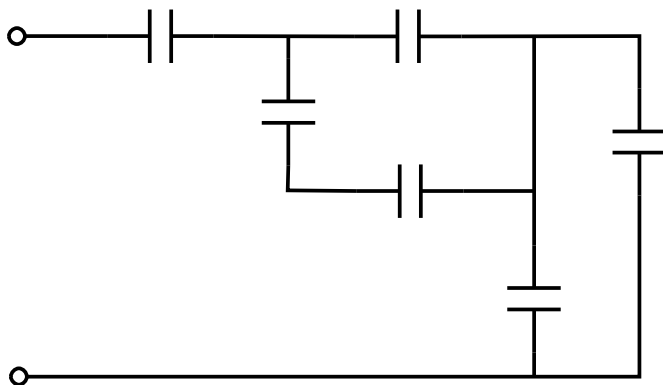


Vraag 2.6 [2]

Bepaal die ekwivalente kapasitansie – aanvaar alle kapasitore het 'n waarde van $10F$.

Question 2.6 [2]

Determine the equivalent capacitance – assume all capacitors have a value of $10F$.

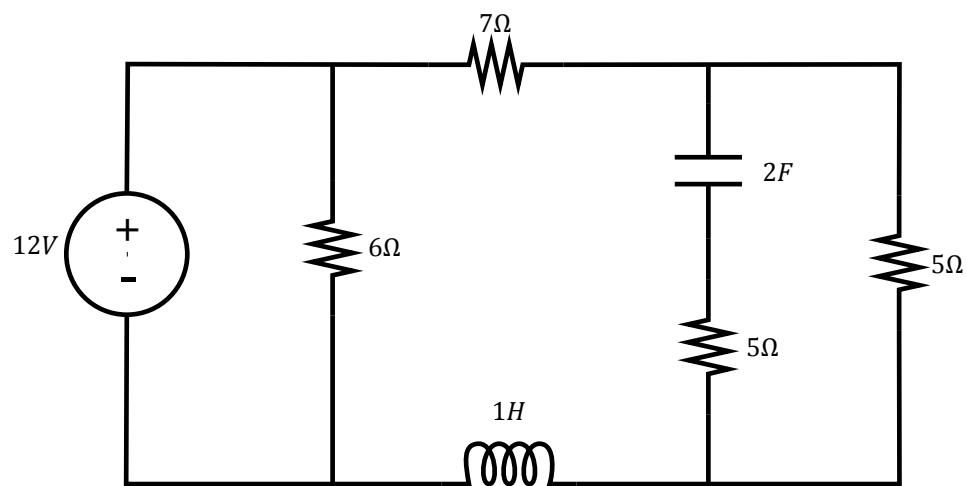


Vraag 2.7 [4]

Aanvaar die baan is in 'n stabiele toestand. Bepaal die gestoorde energie in die induktor en die kapasitor.

Question 2.7 [4]

Assume steady state conditions in the circuit. Determine the energy stored in the inductor and the capacitor.



VRAAG/QUESTION 3 [22]

Vraag 3.1 [6]

a) Gebruik fasor-analise en bepaal $v(t)$:
 $v(t) = 20\cos(\omega t - 60^\circ) - 30\sin(\omega t - 30^\circ)$ A

b) Evalueer die volgende en skryf die antwoord in polêre vorm:

$$\bar{V} = \frac{5\angle 10^\circ - (20 + j5)}{(3 + j4)2\angle 25^\circ}$$

c) Evalueer die determinant en skryf die antwoord in reghoekige form:

$$\begin{vmatrix} 5 + j3 & 2 - j3 \\ 5 & 1 - j \end{vmatrix}$$

Question 3.1 [6]

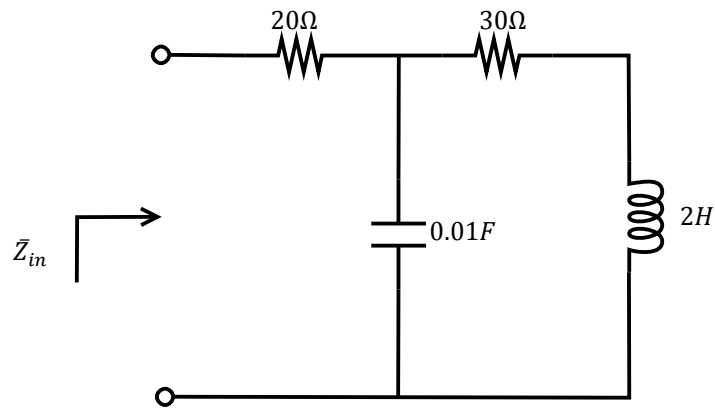
a) Use phasor analysis to find $v(t)$:
 $v(t) = 20\cos(\omega t - 60^\circ) - 30\sin(\omega t - 30^\circ)$ A

b) Evaluate the following and write the answer in polar form:

$$\bar{V} = \frac{5\angle 10^\circ - (20 + j5)}{(3 + j4)2\angle 25^\circ}$$

c) Evaluate the determinant and write the answer in rectangular form:

$$\begin{vmatrix} 5 + j3 & 2 - j3 \\ 5 & 1 - j \end{vmatrix}$$

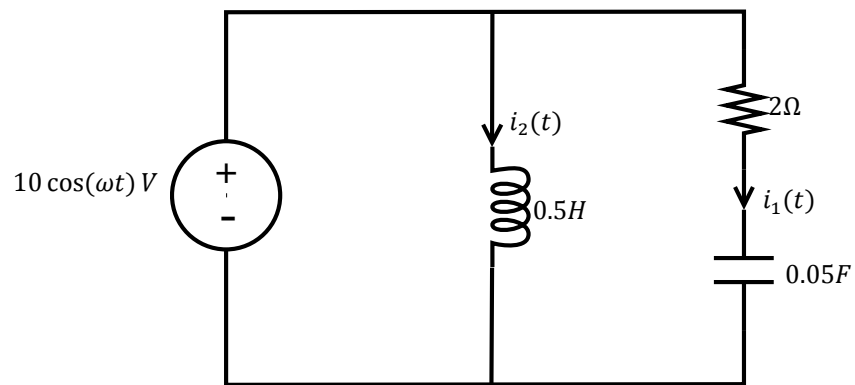
Vraag 3.2 [2]Bepaal die totale impedansie $\bar{Z}_{in}$ as $\omega = 10 \text{ rad/s}$.**Question 3.2 [2]**Determine the total impedance $\bar{Z}_{in}$ if $\omega = 10 \text{ rad/s}$.

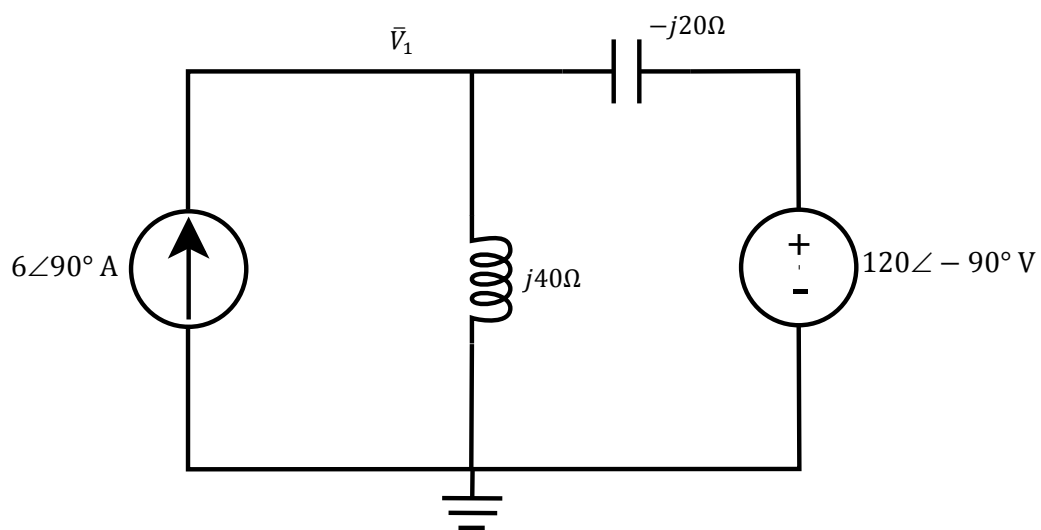
Vraag 3.3 [2]

Vir watter waarde van ω sal die faseverskil tussen $i_1(t)$ en $i_2(t)$ gelyk wees aan 135° . ($\theta_1 - \theta_2 = 135^\circ$)

Question 3.3 [2]

For what value of ω will the phase difference between $i_1(t)$ and $i_2(t)$ be equal to 135° . ($\theta_1 - \theta_2 = 135^\circ$)



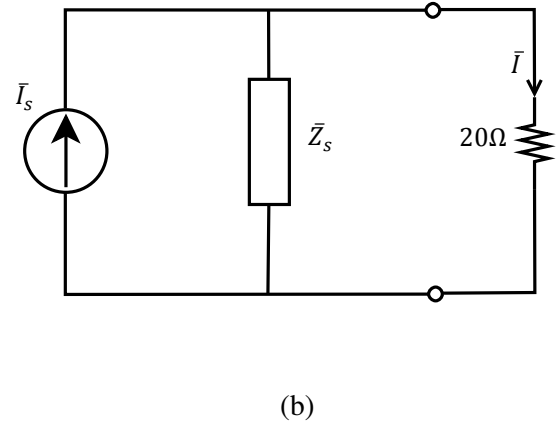
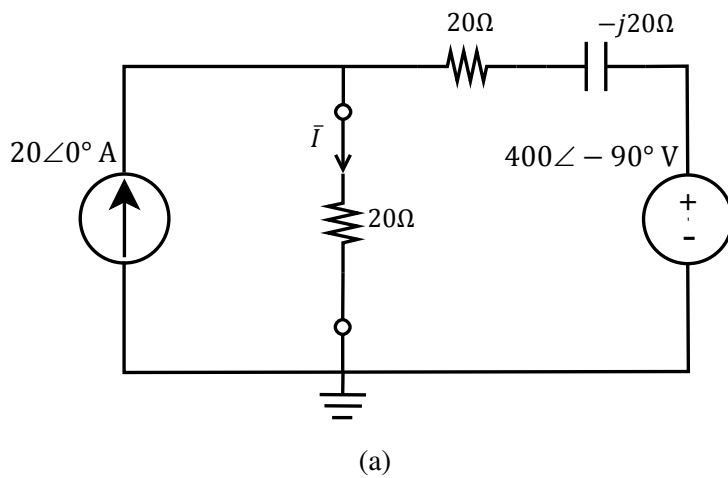
Vraag 3.4 [2]Gebruik node-analise om $\bar{V}_1$ te bereken (in polêre vorm).**Question 3.4 [2]**Use nodal analysis to determine $\bar{V}_1$ (in polar form).

Vraag 3.5 [4]

- a) Sien Figuur (a). Gebruik superposisie om $\bar{I}$ slegs agv die stroombron te bereken (in polêre vorm).
- b) Gebruik brontransformasie om Figuur (a) te transformeer na die stroombaan in Figuur (b) – jy hoef slegs $\bar{I}_s$ (in polêre vorm) te bereken.

Question 3.5 [4]

- a) See Figure (a). Use nodal analysis to determine $\bar{I}$ only due to the current source (in polar form).
- b) Use source transformation to transform the circuit in Figure (a) to that of Figure (b) – you only have to determine $\bar{I}_s$ (in polar form).

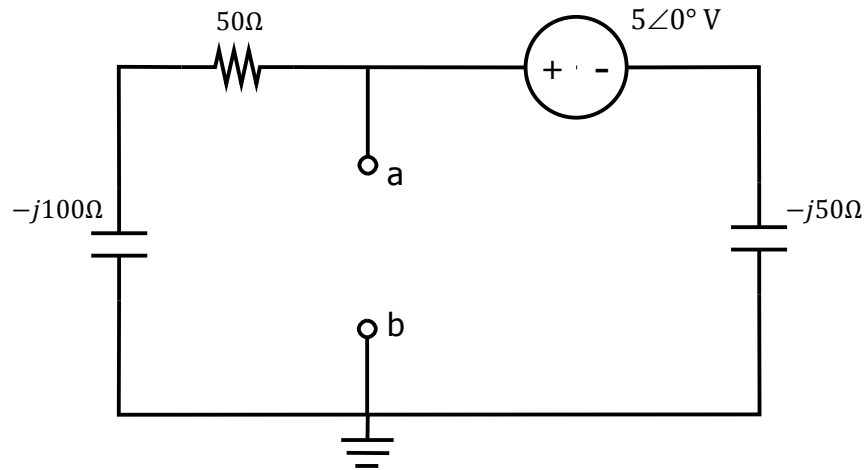


Vraag 3.6 [4]

Bepaal die Norton-ekwivalent tov terminale a-b. Skryf die antwoorde as fasors in polêre vorm.

Question 3.6 [4]

Find the Norton equivalent wrt terminals a-b. Write the answers as phasors in polar form.

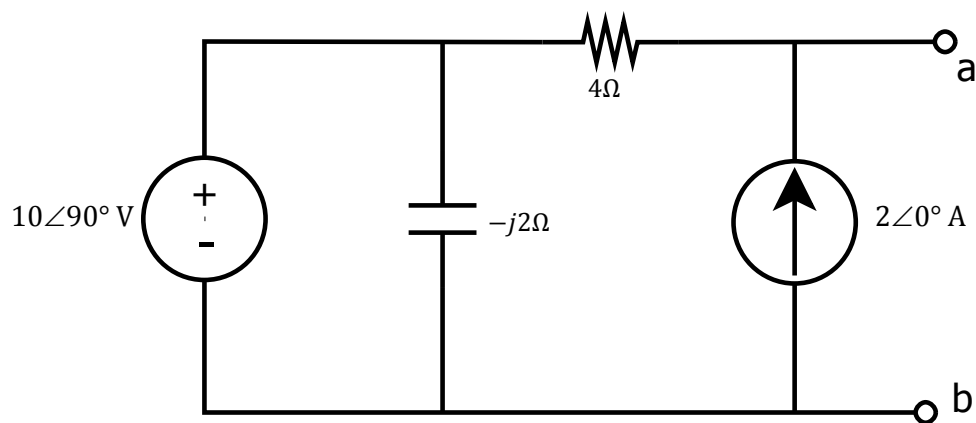


Vraag 3.7 [2]

Bepaal die Thevenin-spanning tov terminale a-b. Skryf die antwoord as 'n fasor in reghoekige vorm.

Question 3.7 [2]

Find the Thevenin voltage wrt terminals a-b. Write the answer as a phasor in rectangular form.



EBN111 EKSAMEN/EXAMINATION (15 JUNIE/JUNE 2015)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:	Memo	Studentenommer: Student number:	<table border="1" style="display: inline-table; border-collapse: collapse;"> <tr> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> <td style="width: 20px; height: 20px;"></td> </tr> </table>								
Voorletters / Initials:		Tel. nr. / Tel no.:									

Q1		Q2		Q3		Tot:	/90
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VRAAG 1 – QUESTION 1 (45)	
1.1	$q = 0.48 \text{ pC}$ ✓
1.2	$i_{(2ms)} = -25 \text{ A}$ ✓✓
1.3	$W = 6.67 \text{ J}$ or $2\frac{2}{3} \text{ J}$ ✓✓
1.4	$\text{Cost} = \text{R } 625$ ✓✓
1.5	$a) P_{(2I_0)} = -8 \text{ W}$ ✓✓ $b) P_{(V_s)} = -6 \text{ W}$ ✓✓
1.6	$\text{Area} = 3.2 \times 10^{-3} \text{ m}^2$ ✓✓
1.7	$a) \text{Nodes} = 4$ ✓ $b) \text{Branches} = 5$ ✓
1.8	$a) R_{eq} = 50 \Omega$ ✓✓ $b) R_{eq} = 50 \Omega$ ✓✓
1.9	$a) V_o = 2 \text{ V}$ ✓ $b) P_{(6\Omega)} = 6 \text{ W}$ ✓✓
1.10	$a = 11$ ✓ $b = -90$ ✓ $c = -1$ ✓ $d = 4$ ✓
1.11	$e = 10$ ✓ $f = 20$ ✓ $g = 1$ ✓ $h = 4$ ✓
1.12	$a) I_C = 127 \mu\text{A}$ ✓✓ $b) I_o = 175 \text{ mA}$ ✓✓
1.13	$V_o = 10 \text{ V}$ ✓✓
1.14	$R = 30 \Omega$ ✓ $I = 3 \text{ A}$ ✓✓
1.15	$I_N = -14 \text{ A}$ ✓✓
1.16	$R = 8.95 \Omega$ ✓✓
1.17	$V_{th} = 100 \text{ V}$ ✓✓

VRAAG 2 – QUESTION 2 (23)

2.1	$v_{01} = -2V$ ✓	$i_0 = 1.5mA$ //
2.2	$v_{02} = 4V$ ✓ (1)	$v_s = -3.75V$ //
2.3	$v_{03} = 1.23V$ //	
2.4	$v_x/v_I = 13.5$ //	$v_x = 4V$ //
2.5	$i_I(2s) = 2000A$ //	$i_2(4s) = -1mA$ //
2.6	$C_{eq} = 4.62F$ //	
2.7	$W_L = 0.5J$ //	$W_C = 25J$ //

VRAAG 3 – QUESTION 3 (22)

3.1	a) $v(t) = 26.46 \cos(\omega t + 19.1^\circ)V$	b) $\bar{V} = 1.56 \angle 117.2^\circ V$
	c) determinant = $-2 + j13$ //	
3.2	$\bar{Z}_{in} = 23 - j11 \Omega$ //	
3.3	$\omega = 10 \text{ rad/s}$ //	
3.4	$\bar{V}_1 = 339.4 \angle -45^\circ V$ //	
3.5	$\bar{I} = 12.65 \angle -18.44^\circ A$ //	$\bar{I}_s = 31.62 \angle -18.45^\circ A$ //
3.6	$\bar{I}_N = 0.1 \angle 90^\circ A$ //	$\bar{Z}_N = 35.36 \angle -81.86^\circ \Omega$ //
3.7	$\bar{V}_{Th} = 8 + j10 V$ //	